NGUYEN VAN ANH

Backend Engineer Intern
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ABOUT ME

Final-year Software Engineering student focused on backend systems - API design, database architecture, and concurrency control. Built and deployed two backend systems, including a ticketing platform verified correct under 50 concurrent requests with zero oversell. Seeking a backend internship to apply these skills on a real product.

PROJECTS

IT Asset Management System | GitHub: iamnvanh205/itam-system | Demo: itam.vananh.io.vn
[05 - 07/2026]
Java, Spring Boot, React, TypeScript, PostgreSQL, JWT, Docker, JUnit 5, Mockito, Testcontainers
- Built a full IT asset lifecycle system (purchase → assignment → maintenance → audit → disposal) with true multi-branch operation (branch-level data isolation), replacing manual Excel tracking for 50-100 employees / 200-300 assets.
- Designed a 3-step approval workflow (Employee → Manager → IT Staff) with auto-approve thresholds and a full state machine, backed by transactional side-effects and rollback on failure.
- Built a custom JWT auth system with refresh token rotation (token family/generation model, reuse detection aligned with RFC 9700), supporting multi-device sessions and instant revocation; RBAC across 4 roles via Spring Security.
- Extended the system with a Software License Management module (entitlements, seat assignment by user/device, row-level locking to prevent over-allocation) and QR-based asset audits with automatic discrepancy detection.
- Covered business logic with 120 unit/integration tests (JUnit 5, Mockito, Testcontainers), 61.6% line coverage; full CI/CD via GitHub Actions — Docker build → GHCR → auto-deploy.

Event Ticketing System | GitHub: iamnvanh205/event-ticketing | Demo: event.vananh.io.vn
[07/2026 - Present]
Java, Spring Boot, React, TypeScript, PostgreSQL, WebSocket/STOMP, JWT/OAuth2, Docker
- Built an event ticketing platform to solve a specific hard problem: preventing ticket oversell and duplicate check-ins when many users compete for the same limited tickets.
- Designed a reserve-then-confirm booking flow using DB-level pessimistic locking (SELECT FOR UPDATE) to serialize concurrent purchase attempts on the last ticket.
- Validated correctness with concurrency tests: 50 concurrent requests competing for the last ticket, zero oversell, zero duplicate reservation.
- Built a real-time organizer dashboard (WebSocket/STOMP) and QR-based check-in flow, with role-based access across 4 roles (Admin, Organizer, Check-in Staff, Customer).
- Implemented JWT-based authentication via OAuth2, with refresh token handling and logout support through token invalidation.
- Self-hosted the full stack on a VPS using Docker Compose, Nginx reverse proxy, and HTTPS, with the same CI/CD pipeline as ITAM: Docker build → push to GHCR → auto-deploy via GitHub Actions

EDUCATION

Hanoi Open University | 2023 – 2027
Faculty of Information Technology - Major: Software Engineering

SKILLS

- Languages: Java, JavaScript, TypeScript, SQL
- Frameworks: Spring Boot, Spring Security, Hibernate, JPA
- DevOps: Docker, Git, GitHub Actions, Linux
- Architecture: REST API, Microservices, Clean Architecture, SOLID
- Spoken Languages: Vietnamese (Native), English (proficient in reading technical documentation)